

## AMATH 582: HOMEWORK 2

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**ABSTRACT.** This report describes the process of designing an algorithm to characterize the movements of a robot, OptimuS-VD, in real-time of 3 movements: jumping, running, and walking. The algorithm uses Principle Component Analysis as an unsupervised learning method to reduce the dimensions of the data. Then, centroid-based classification is used to classify the activity for each sample dataset. Training data is used to train the Principle Component analysis model, then test data is projected onto the k-space created. With an optimal number of spatial nodes of 2, this model is 88% accurate at classifying the training data and 98% accurate at classifying the test data. Additionally, Principle Component Analysis can be paired with a different supervised model that uses K-Nearest Neighbors. Using 2 spatial modes and an optimal K-nearest neighbor number of 6 neighbors, this model is 99% accurate at classifying the training data and 100% accurate at classifying the test data.

### 1. INTRODUCTION AND OVERVIEW

This report describes the process of designing an algorithm to characterize the movements of a robot, OptimuS-VD, in real-time. Data were provided of recorded movements of 38 of the robot's joints at 60 Hz, collected as Euler angles, which can be transformed to xyz coordinates. The samples were recorded for 1.4 seconds, causing each sample to be represented as a matrix comparing the 114 features (due to three different axes of 38 data points) to the 100 time steps. Five samples were taken of 3 different movements: walking, jumping, and running. The goal of this process is to build a projection of the recordings to a lower dimension and then the number of coordinates to visualize the movements. So, Principle Component Analysis (PCA), a supervised learning concept, was used for this reduction and projection. The algorithm that determined the activity of a sample dataset was designed with unsupervised learning as centroids of each activity were found using the provided activity type of each training sample. Minimal distance of a data point from each activity's centroid classified the activity type of each data point. The provided sample datasets were used to train the model, and additional provided datasets tested the model. The accuracies of these classifications evaluated the performance of this model.

### 2. THEORETICAL BACKGROUND

A large part of this model depends on Principle Component Analysis (PCA) to reduce the dimensions of the datasets. The goal of Principle Component Analysis is to find a sequence of orthonormal basis functions so that the first term gives the best 1-term approximation of the function, the first and second gives the best 2-term approximation, and so on. These orthonormal functions are called the Proper Orthogonal Modes (POD). The following formula demonstrates the POD expansion of a function, where  $f(x, t)$  is the original function,  $N$  is the total number of modes,  $a_j(t)$  determines the weights of the spatial nodes, and  $\phi_j(x)$  are the modes [4].

$$f(x, t) \approx \sum_{j=1}^N a_j(t) \phi_j(x)$$

Singular Value Decomposition (SVD) is the basis of this method because because its outcome "orders singular values and associated orthonormal basis functions that capture as much energy as possible." SVD

works through the following formula, where  $U$  and  $V$  are orthonormal bases and  $\Sigma$  contains the singular values of the matrix  $A$  [4].

$$A = U\Sigma V^*$$

Data can be projected onto truncated Principle Component modes to reduce its dimensionality, reducing noise and maintaining key variations. The equation used to project data onto this truncated K-space is shown below, where  $U_K^T X$  is the basis of the truncated K-space,  $X$  is the original data, and  $\tilde{X}_{PC_K}$  is the projected data in the K-space.

$$U_K^T X = \tilde{X}_{PC_K}$$

To evaluate the amount of energy that is retained after the data is projected onto the truncated K-space, the percent of total energy is calculated using the following equation, where  $E_m$  is the cumulative sum of the normalized energy,  $m$  is the number of modes in the truncated K-space,  $N$  is the total number of modes, and  $\sigma_i$  is the singular value for each mode.

$$E_m = \frac{\sum_{i=1}^m \sigma_i^2}{\sum_{i=1}^N \sigma_i^2}$$

The main method in this report uses supervised learning by comparing the distance between the dataset and the classification centroids, commonly called centroid-based classification. This method takes training data and defines a centroid for each classification type. Classification is established by evaluating the distance between test data points and the centroid. Centroid-based classification has been used for a variety of applications, including many movement-based application. One research paper used this method to recognize people based on features of their walking form, and they had a high classification accuracy of 98% for the favorable aspect angle, showing high performance of this method [1].

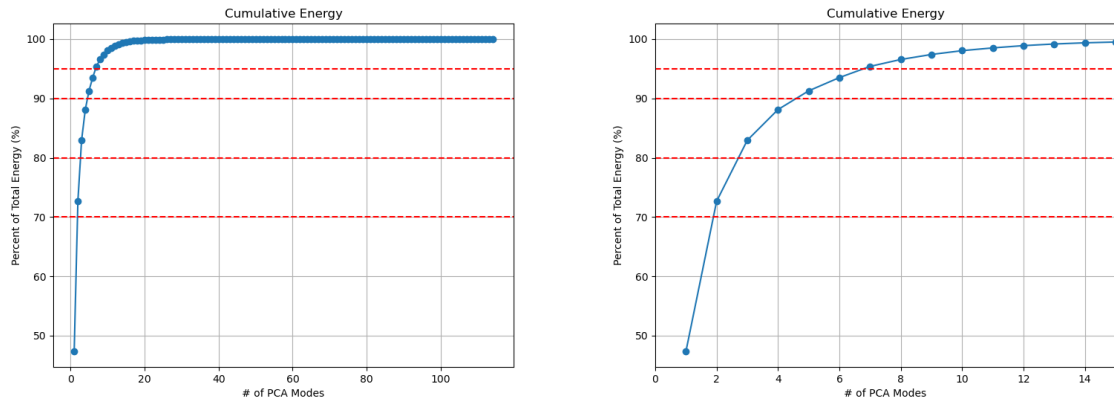
### 3. ALGORITHM IMPLEMENTATION AND DEVELOPMENT

The primary function used to analyze this dataset is Scikit Learn's PCA Decomposition. This function uses SVD to project data to a lower space and centers the data [5]. Scikit Learn's Metrics was also used to calculate the accuracy score of the training and test data [5]. The last Scikit Learn package used was their Neighbors package, which was used to perform K-Nearest Neighbors classification [6]. This function the k-nearest neighbors vote to classify data by comparing the dataset to its closest neighbors. Additionally, NumPy was used to manage matrices [2]. To plot everything, Matplotlib's PyPlot was used [3].

### 4. COMPUTATIONAL RESULTS

A training parameter was entered into the PCA algorithm that contained all the sample data. The shape of this parameter was 114 x 1500, maintaining the convention that the rows represent the features and spatial variables while the columns corresponded to the samples and time steps. This training parameter was transposed and entered into PCA to truncate the matrix. To determine the number of k spatial nodes needed to maintain adequate cumulative energy to represent the data, a range of 1 to 114 spatial nodes were entered into the PCA algorithm. The cumulative energy was plotted, and the full plot is shown in Figure 1a, and a zoomed in plot is shown in Figure 1b. As shown in the plot, 70% energy is achieved with 2 node, 80% energy with 3 nodes, 90% energy after 3 nodes, and 95% with 5 nodes.

The first 2 principle components were plotted against each other to show how they display the data of different activities. By doing this, the orthogonal directions of maximum variance and second maximum variances are plotted against each other, showing the movement of each activity. This is shown in Figure 2. To create a model that identifies the sample's activity with only the dataset, a vector of ground truth labels was established to calculate the centroid for each activity. Each activity's centroid was calculated as the mean of the 5 PCA-applied sample datasets for each activity. The mean was taken for each principle component direction, so each centroid had k-dimensions. The centroids in 2-dimensions are plotted in Figure 2 as well. In this K-space, the walking and running data points are very separate from the other with no overlap of data, and the centroids are very far apart as well. There is some overlap between the walking and jumping data points, but the majority of the data are separate, and the centroids are away from each other. Overall, the activities are distinct from each other in this K-space.



(A) Full Plot of the number PCA Modes vs. Cumulative Energy (B) Zoomed-In Plot of the number of PCA Modes (0-15) vs. Cumulative Energy

FIGURE 1. These plots show the relationship between the number of PCA Modes and cumulative energy, demonstrating the amount of energy of the full size matrix was kept as PCA reduced the matrix to different K numbers of PCA modes. There are horizontal dashed lines at 70%, 80%, 90%, and 95% to assist with finding the number of PCA modes that achieve these energy percentages.

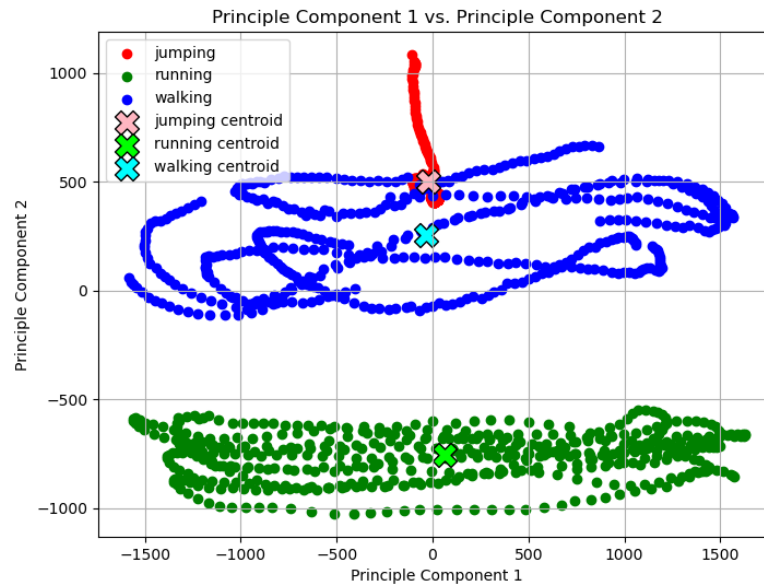


FIGURE 2. This plot shows the movements of each activity (jumping, running, walking) in 2 dimensions by plotting Principle Component 1 (the direction with the largest variance) vs. Principle Component 2 (the direction with the second largest variance). The result shows 3 distinct movements between the 3 activities. The centroids of each movement are also shown.

To show how adding another principle component affects how the data is displayed, the first 3 principle components were also plotted against each other. This plot is shown in Figure 3 with the centroids of each activity. In this K-space, the activities are still distinct from each other, but there is more overlap than there

was in only 2 dimensions. Each activity's data points seem to take up different corners of the 3-D space, but some of the data points are overlapping, causing the centroids to be closer to data points of other activities. Because of this, centroid-differencing methods will likely perform better with 2 spatial modes than 3 spatial modes.

Principle Component 1 vs. Principle Component 2 vs. Principle Component 3

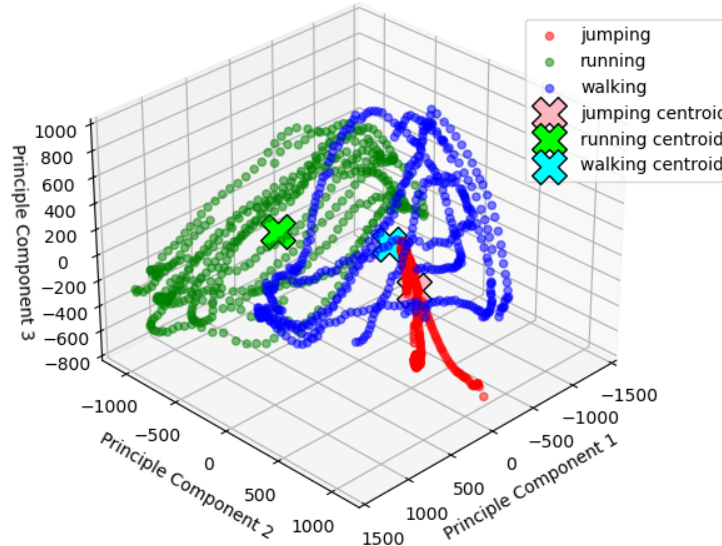
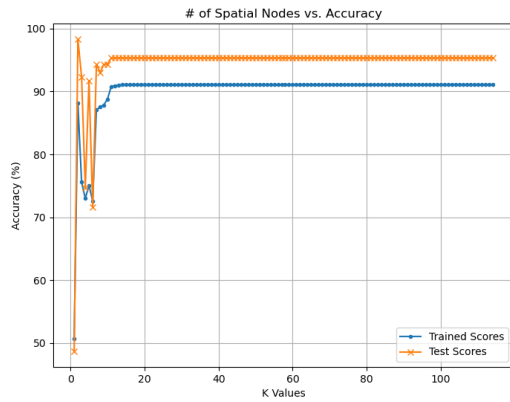


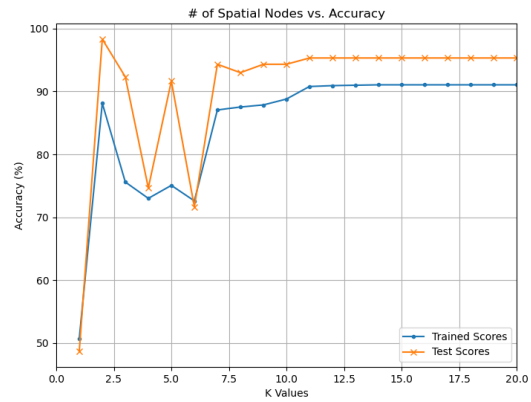
FIGURE 3. This plot shows the movements of each activity (jumping, running, walking) in 3 dimensions by plotting Principle Component 1 (the direction with the largest variance) vs. Principle Component 2 (the direction with the second largest variance) vs. Principle Component 3 (the direction with the third largest variance). The result shows 3 distinct movements between the three activities. The centroids of each movement are also shown.

For each data point, each activity's distance was found as the Frobenius norm of the data point in  $k$ -space minus the activity's centroid. The minimal activity distance classified each data point as that activity. The model's activity classifications of the training data were compared to their known activity classifications as an accuracy percentage. The testing data was projected on the same  $k$ -space that was fitted to the training data, and this centroid-difference method was also used to get a test data accuracy score. Figure 4a has the plot of the relationship between the  $K$  values (number of spatial nodes) from 1 to 115 compared to the accuracy. Figure 4b shows a closer view of this relationship with  $K$  values from 1 to 20. The number of spatial modes ( $K$ ) that achieves peak accuracy for both training and test values is 2, where the training score is about 88% and the test score is about 98%. The test score being significantly higher than the training score shows that the data is being under-fit, likely due to a small amount of data.

An alternative classifier, K-Nearest Neighbors (KNN) classifier, was also used on these samples and its performance was evaluated similarly. Instead of using centroids to determine the activity of each sample, the KNN classifier computes the distance between a test point and its  $K$ -nearest points. The training data was transformed to the 2-dimensional space using PCA then used to fit the KNN model. The test data was also transformed to this space through PCA, then the accuracy score was computed for the training and test data by using the score function to determine the percentage of correct labels. The relationship between the number of K-Neighbors ( $K$ ) and the accuracy is plotted in Figure 5. The training score reaches 100% and is highest with  $K=1$  neighbor, but then slowly decreases. The test score reaches 100% accuracy with  $K=6$  neighbors and remains at this score. This shows that the data is still being under-fit. The optimal  $K$  value



(A) Full Plot of the K Value vs. Accuracy



(B) Zoomed-In Plot of the K Value vs. Accuracy

FIGURE 4. These plots show the relationship between the K value (number of spatial nodes) and the accuracy the model's classification of the training samples' activities and the testing samples' activities.

is 6, where the training score is about 99% and the test score is 100%. Compared to the centroid-difference model, these accuracy scores are a couple percentages higher, showing that KNN is a more accurate model.

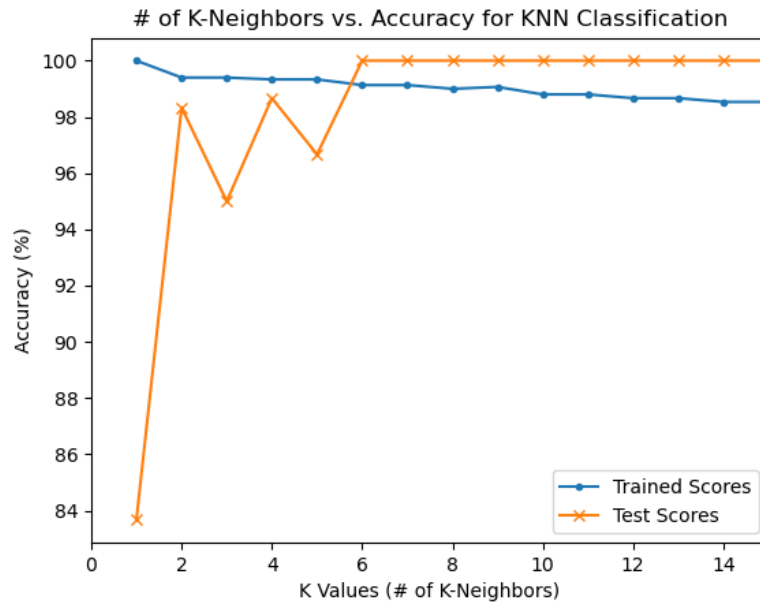


FIGURE 5. These plots show the relationship between the K value (number of K-neighbors) and the accuracy of the KNN model's classification of the training samples' activities and the testing samples' activities.

## 5. SUMMARY AND CONCLUSIONS

This report showed that a supervised model that depends on centroid distances paired with an unsupervised model the uses Principle Component Analysis can accurately classify the activities performed by OptimuS-VD in each sample. With an optimal number of spatial nodes of 2, this model is 88% accurate

at classifying the training data and 98% accurate at classifying the test data. Additionally, Principle Component Analysis can be paired with a different supervised model that uses K-Nearest Neighbors. Using 2 spatial modes and an optimal K-nearest neighbor number of 6 neighbors, this model is 99% accurate at classifying the training data and 100% accurate at classifying the test data.

To further extend this investigation, more test samples should be analyzed by these models. Each jumping, running, and walking sample all seemed relatively similar to other samples of the same activity, which made it easier for these models to accurately classify the data. Different styles of jumping, running, and walking should be provided to these models to ensure that they can accurately classify a larger variety of each activity.

#### ACKNOWLEDGEMENTS

The TA, Saba Heravi, provided a lot of assistance for the report through her office hours. This support included verifying matrix sizes and verifying graphs.

Additionally, I received help from many peers in this course, with regard to verifying graphs, debugging, and discussing how to use new methods: Anja Vogt, Anna Rutherford, Jonathan Elsner, Maddy Kovaleski, Hailey Sparks, Jaxon Tuggle, Hunter Lybert, and John Murphy.

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